

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid1, pid2;

    /* Create pipe */
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(EXIT_FAILURE);
    }

    /* Create first child */
    pid1 = fork();

    if (pid1 < 0)
    {
        perror("fork");
        exit(EXIT_FAILURE);
    }

    if (pid1 == 0)
    {
        /* First child: execute ls -l */

        close(pipefd[0]); // Close unused read end

        /* Redirect stdout to pipe */
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", "-l", (char *)NULL);

        perror("execlp ls");
        exit(EXIT_FAILURE);
    }

    /* Create second child */
    pid2 = fork();

    if (pid2 < 0)
    {
        perror("fork");
        exit(EXIT_FAILURE);
    }
}
```

```

if (pid2 == 0)
{
    /* Second child: execute grep ".c" */

    close(pipefd[1]); // Close unused write end

    /* Redirect stdin from pipe */
    dup2(pipefd[0], STDIN_FILENO);

    close(pipefd[0]);

    execlp("grep", "grep", ".c", (char *)NULL);

    perror("execlp grep");
    exit(EXIT_FAILURE);
}

/* Parent process */
close(pipefd[0]);
close(pipefd[1]);

/* Wait for both children */
waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);

return 0;
}

```

Screen Shots:

```

sohan_jadala@DESKTOP-OI7F:~$ ls
2520080060_OSSP  Fork  OSLab  OSSP_Lab  Practical  Practice  Skill
sohan_jadala@DESKTOP-OI7F:~$ cd 2520080060_OSSP
-bash: cd: 2520080060_OSSP: No such file or directory
sohan_jadala@DESKTOP-OI7F:~$ cd 2520080060_OSSP
sohan_jadala@DESKTOP-OI7F:~/2520080060_OSSP$ ls
2520080060_ForgeOS  2520080060_Practical  2520080060_Skill  OS_CO-1_Practice_Programmes.pdf
sohan_jadala@DESKTOP-OI7F:~/2520080060_OSSP$ cd 2520080060_Practical
sohan_jadala@DESKTOP-OI7F:~/2520080060_OSSP/2520080060_Practical$ gedit Practical_5.c

** (gedit:564): WARNING **: 08:26:03.107: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:564): WARNING **: 08:26:03.107: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
sohan_jadala@DESKTOP-OI7F:~/2520080060_OSSP/2520080060_Practical$ gcc Practical_5.c -o Practical_5
sohan_jadala@DESKTOP-OI7F:~/2520080060_OSSP/2520080060_Practical$ ./Practical_5
Consumer: Received 100000 integers

Producer: Sent 100000 integers
Communication time: 0.065745 seconds
Communication rate: 1521024.10 integers/sec
sohan_jadala@DESKTOP-OI7F:~/2520080060_OSSP/2520080060_Practical$ ls -l | grep ".c"
grep.c: command not found
sohan_jadala@DESKTOP-OI7F:~/2520080060_OSSP/2520080060_Practical$ ls -l | grep ".c"
drwxr-xr-x 2 sohan_jadala sohan_jadala 4096 Aug  8 08:21 Practical-01
-rwxr-xr-x 1 sohan_jadala sohan_jadala 16344 Aug  8 09:25 Practical3
-rw-r--r-- 1 sohan_jadala sohan_jadala 1162 Aug  8 09:24 Practical3.c
-rwxr-xr-x 1 sohan_jadala sohan_jadala 16296 Aug 14 08:55 Practical4
-rw-r--r-- 1 sohan_jadala sohan_jadala 835 Aug 14 08:54 Practical4.c
-rwxr-xr-x 1 sohan_jadala sohan_jadala 16392 Sep 12 08:28 Practical_5
-rw-r--r-- 1 sohan_jadala sohan_jadala 1647 Sep 12 08:27 Practical_5.c
-rwxr-xr-x 1 sohan_jadala sohan_jadala 16048 Aug 29 10:07 create_FIFO
-rw-r--r-- 1 sohan_jadala sohan_jadala 218 Aug 29 10:03 create_FIFO.c
-rw-r--r-- 1 sohan_jadala sohan_jadala 254 Aug 17 05:21 file_permission.c
-rw-r--r-- 1 sohan_jadala sohan_jadala 265 Aug 20 10:03 pipe_read.c
-rw-r--r-- 1 sohan_jadala sohan_jadala 428 Aug 20 10:17 pipe_write.c
-rw-r--r-- 1 sohan_jadala sohan_jadala 124 Aug 20 17:00 signal_2.c
-rw-r--r-- 1 sohan_jadala sohan_jadala 239 Aug 20 15:39 signals_1.c
sohan_jadala@DESKTOP-OI7F:~/2520080060_OSSP/2520080060_Practical$

```

